

# Yunjin (Vivian) Yu

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## Experiences

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### Sonic Arts R&D, Qualcomm Institute

*July 2026 - Present*

*Undergraduate Researcher under Professor Shahrokh Yadegari*

- Working on the Space3D project, a **real-time ray-tracing spatial audio rendering engine**, integrating the optimized HRTF interpolation pipeline and beginning work on an existing lab model for **ML-based HRTF personalization**.
- Optimized the lab's existing SOFA preprocessing codebase to **auto-orient files** and **map measurements** onto a consistent coordinate system before HRIR truncation, improving the interpolation pipeline's runtime efficiency by **~6x**.
- Generalized the lab's per-database triangle-mesh builders into a single **KNN-based builder**, extending interpolation support to any SOFA file rather than four hardcoded databases.

### ECE Makerspace, UCSD Jacobs School of Engineering

*September 2026 - Present*

*Makerspace Staff*

- Maintaining **fabrication and electronics equipment** (3D printers, laser cutters, soldering stations), tracking inventory, and training students and faculty on safe operation and troubleshooting.

## Projects

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### Theramen - Contactless Synthesizer with Multi-Device I2C Coordination

<https://github.com/vivianyunjinyu/theramen>

- Developed **embedded C++ firmware** on an STM32H750 microcontroller, implementing **driver logic** for two I2C time-of-flight sensors alongside real-time audio synthesis and OLED display output.
- Debugged an I2C address collision between identical sensors via hardware reset sequencing, then optimized I2C bus utilization by batching display writes into single burst transactions to sustain real-time sensor polling.

### Vocoduh - FFT Channel Vocoder

<https://github.com/vivianyunjinyu/FFT-Vocoder-Plugin>

- Developed a **real-time audio signal processing algorithm** (FFT/STFT-based channel vocoder) in C++, replacing a fixed filter-bank architecture with finer frequency-bin decomposition for improved frequency resolution.
- Implemented tunable audio processing controls (frequency-weighted noise injection, spectral whitening) to improve signal clarity, enabling user-adjustable audio quality tuning.

### Whoosh - Tempo-Synced LFO Tool

<https://github.com/vivianyunjinyu/Whoosh-Plugin>

- Developed a computationally efficient **DSP lookup-table cache** with dirty-flag regeneration, reducing runtime cost by replacing per-sample power-function evaluation with interpolated lookups.
- Synchronized real-time audio processing to host DAW tempo, eliminating dependency on external audio input.

## Skills

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**Programming:** Python (PyTorch, TensorFlow, scikit-learn, Pandas), C++ (Embedded), Java, MATLAB, Arduino

**Embedded Sys and SBCs:** Arduino, ESP32, STM32, Electrosmith Daisy, Raspberry Pi

**Tools & Formats:** Git/GitHub, Linux, SSH, LTSpice, Onshape, PureData

**Lab Equipment:** Keysight Oscilloscope, Multimeter, Power Supply, Waveform Generator, Soldering Tools

**Audio/Video:** AVID Pro Tools (Certified Specialist), Ableton Live, Adobe Premiere, DaVinci Resolve

**General:** Microsoft Office (Word, Excel, PowerPoint), Google Workspace, Canva

## Education

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### University of California, San Diego

*Expected Graduation: June 2027*

### Interdisciplinary Computing and the Arts, minor in Electrical Engineering

- Coursework: Digital Signal Processing, Machine Learning for Music, Linear Systems, Circuits and Systems